

Log Analysis with CLI Tools

Outcomes

By the end of this lab, you will be able to:

- Use *grep* to search for specific patterns in log files
- Utilize *awk* to extract and manipulate specific fields from log data
- Employ *sed* to perform basic text transformations on log entries
- Chain command-line tools together to perform complex log analysis tasks
- Identify suspicious activity in web server logs
- Create professional log analysis reports

Objective

Master essential Linux command-line tools (*grep*, *awk*, *sed*, *cut*, *sort*, *uniq*) to parse and analyze log files for security investigations

Prerequisites

- Linux system (Ubuntu VM, or Kali Linux)
- Basic understanding of Apache log format
- Text editor (nano, vim, or VS Code)

Part 1: Understanding Apache Log Format (15 minutes)

Step 1: Apache Combined Log Format

```
LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\"" combined
```

Step 2: Download Sample Log File

Create a working directory:

```
mkdir -p ~/soc-labs/week3
cd ~/soc-labs/week3
```

```
touch@ubuntu:~$ sudo su
root@ubuntu:/home/touch# mkdir -p ~/soc-labs/wazhu
root@ubuntu:/home/touch# cd ~/soc-labs/wazhu
root@ubuntu:~/soc-labs/wazhu# wget https://raw.githubusercontent.com/elastic/examples/master/Common%20Data%20Formats/apache_logs/apache_logs -O access.log
--2026-05-17 05:11:00-- https://raw.githubusercontent.com/elastic/examples/master/Common%20Data%20Formats/apache_logs/apache_logs
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2370789 (2.3M) [text/plain]
Saving to: 'access.log'

access.log          40%[=====>] 944.00K  45.6KB/s  eta 38s
```

Download sample log file:

```
cp /path/to/soc-training-program/Lab-Resources/Sample-Data/apache-access.log .
```

or,

```
wget https://raw.githubusercontent.com/elastic/examples/master/Common%20Data%20Formats/apache_logs/apache_logs -O access.log
```

```
root@ubuntu:~/soc-labs/wazhu# wget https://raw.githubusercontent.com/elastic/examples/master/Common%20Data%20Formats/apache_logs/apache_logs -O access.log
--2026-05-17 05:11:00-- https://raw.githubusercontent.com/elastic/examples/master/Common%20Data%20Formats/apache_logs/apache_logs
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2370789 (2.3M) [text/plain]
Saving to: 'access.log'

access.log          40%[=====>] 944.00K  45.6KB/s  eta 38s
```

Verify the file:

```
ls -lh access.log
head -5 access.log
```

```

2026-05-17 05:12:04 (37.5 KB/s) - 'access.log' saved [2370789/2370789]

root@ubuntu:~/soc-labs/wazhu# ls -lh access.log
-rw-r--r-- 1 root root 2.3M May 17 05:12 access.log
root@ubuntu:~/soc-labs/wazhu# head -5 access.log
83.149.9.216 - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:43 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard3.png HTTP/1.1" 200 171717 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:47 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 26185 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:12 +0000] "GET /presentations/logstash-monitorama-2013/plugin/zoom-js/zoom.js HTTP/1.1" 200 7697 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:07 +0000] "GET /presentations/logstash-monitorama-2013/plugin/notes/notes.js HTTP/1.1" 200 2892 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
root@ubuntu:~/soc-labs/wazhu#

```

Part 2: Basic Log Analysis Commands

1: Count Total Requests

```
wc -l access.log
```

```

root@ubuntu:~/soc-labs/wazhu# wc -l access.log
10000 access.log

```

2: Extract IP Addresses

Extract just the IP addresses (first field):

```
awk '{print $1}' access.log | head -10
```

```

root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | head -10
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
83.149.9.216
root@ubuntu:~/soc-labs/wazhu#

```

Sort the IP addresses:

```
awk '{print $1}' access.log | sort | head -10
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | sort | head -10
100.2.4.116
100.2.4.116
100.2.4.116
100.2.4.116
100.2.4.116
100.2.4.116
100.43.83.137
100.43.83.137
100.43.83.137
100.43.83.137
root@ubuntu:~/soc-labs/wazhu#
```

Remove duplicates:

```
awk '{print $1}' access.log | sort | uniq | head -10
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | sort | uniq | head -10
100.2.4.116
100.43.83.137
101.119.18.35
101.199.108.50
101.226.168.196
101.226.168.198
101.226.33.222
103.245.44.13
103.247.192.5
103.25.13.22
root@ubuntu:~/soc-labs/wazhu#
```

Count unique IPs:

```
awk '{print $1}' access.log | sort | uniq | wc -l
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | sort | uniq | wc -l
1753
root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | sort -u | wc -l
1753
```

3: Find Top Talkers

```
awk '{print $1}' access.log | sort | uniq -c | sort -rn | head -10
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | sort | uniq -c | sort -rn | head -10
482 66.249.73.135
364 46.105.14.53
357 130.237.218.86
273 75.97.9.59
113 50.16.19.13
102 209.85.238.199
99 68.180.224.225
84 100.43.83.137
83 208.115.111.72
82 198.46.149.143
root@ubuntu:~/soc-labs/wazhu#
```

4: Analyze HTTP Status Codes

```
grep " 404 " access.log | wc -l
```

```
82 198.46.149.143
root@ubuntu:~/soc-labs/wazhu# grep " 404 " access.log | wc -l
213
root@ubuntu:~/soc-labs/wazhu#
```

5. Identify Suspicious URLs

```
awk '$9 == 404 {print $7}' access.log | sort | uniq -c | sort
-rn | head -5
```

```
root@ubuntu:~/soc-labs/wazhu# awk '$9 == 404 {print $7}' access.log | sort | uniq -c | sort -rn | head -5
61 /files/logstash/logstash-1.3.2-monolithic.jar
32 /presentations/logstash-puppetconf-2012/images/office-space-printer-beat-down-gif.gif
6 /wp/wp-admin/
6 /wp-login.php?action=register
6 /wp-login.php
root@ubuntu:~/soc-labs/wazhu#
```

Part 3: Advanced Pattern Matching with grep (45 minutes)

6: Detect Scanning Activity

```
grep -i "nikto" access.log | wc -l
```

Get unique IPs using Nikto:

```
grep -i "nikto" access.log | awk '{print $1}' | sort -u
```

```
root@ubuntu:~/soc-labs/wazhu# grep -i "nikto" access.log | wc -l
0
root@ubuntu:~/soc-labs/wazhu# grep -i "nikto" access.log | awk '{print $1}' | sort -u
root@ubuntu:~/soc-labs/wazhu#
```

7: Detect SQL Injection Attempts

Common SQL injection patterns:

- ' OR '1'='1
- UNION SELECT
- DROP TABLE
- ; -- (SQL comment)
- %27 (URL-encoded single quote)

Search for SQL keywords:

```
grep -iE "(union|select|insert|update|delete|drop|exec|scrip
t)" access.log | wc -l
```

```
root@ubuntu:~/soc-labs/wazuh# grep -iE "(union|select|insert|update|delete|drop|exec|script)" access.log | wc -l
124
root@ubuntu:~/soc-labs/wazuh#
```

Search for encoded SQL:

```
grep -E "(%27|%20union|%20select)" access.log
```

[illegible]

Extract suspicious requests:

```
grep -iE "(union|select)" access.log | awk '{print $1, $7}' | head -108: Detect XSS Attempts
```

8: Detect XSS Attempts

XSS patterns:

- `<script>`

- javascript:
- onerror=
- %3Cscript%3E (URL-encoded)

```
grep -iE "(<script|javascript:|onerror=|%3Cscript)" access.log
```

```
root@ubuntu:~/soc-labs/wazhu# grep -iE "(union|select)" access.log | awk '{pr
int $1}' | sort | uniq -c | sort -rn
root@ubuntu:~/soc-labs/wazhu# grep -iE "(<script|javascript:|onerror=|%3Cscri
pt)" access.log
root@ubuntu:~/soc-labs/wazhu# grep -iE "(<script|javascript:|onerror=|%3Cscri
pt)" access.log | wc -l
0
root@ubuntu:~/soc-labs/wazhu# grep -iE "(<script|javascript:|onerror=)" acces
s.log | awk '{print $1}' | sort | uniq -c | sort -rn
root@ubuntu:~/soc-labs/wazhu#
```

9: Detect Directory Traversal

```
grep -E "(\.\.\/|\.\.%.2[Ff])" access.log
or,
grep -E "(\.\.\/|\.\.%.2[Ff])" access.log | wc -l
```

Part 4: Advanced Analysis with awk

10: Analyze Traffic by Time

```
awk '{print $4}' access.log | cut -d: -f2 | sort | uniq -c |
sort -rn
```

```

root@ubuntu:~/soc-labs/wazhu# grep -E "(\\.\\.\\.\\.\\.\\.%2[FE])" access.log | wc -l
0
root@ubuntu:~/soc-labs/wazhu# awk '{print $4}' access.log | cut -d: -f2 | sort | uniq -c | sort -rn
498 14
496 15
493 19
486 20
484 17
478 18
475 13
473 16
462 12
459 11
453 21
443 10
371 05
366 06
365 02
364 09
361 00
360 01
357 07
356 23
355 04
354 03
346 22
345 08

```

11: Analyze Request Methods

```

awk '{print $6}' access.log | tr -d '"' | sort | uniq -c | sort -rn

```

```

root@ubuntu:~/soc-labs/wazhu# awk '{print $6}' access.log | tr -d '"' | sort | uniq -c | sort -rn
9952 GET
42 HEAD
5 POST
1 OPTIONS
root@ubuntu:~/soc-labs/wazhu#

```

12: Extract POST Requests

```

awk '$6 == "\"POST\""' access.log | awk '{print $1}' | sort -u

```

```

root@ubuntu:~/soc-labs/wazhu# awk '$6 == "\"POST\""' access.log | awk '{print $1}' | sort -u
37.115.186.244
78.173.140.106
91.236.74.121
root@ubuntu:~/soc-labs/wazhu#

```

Get POST request details:

```

awk '$6 == "\"POST\""' {print $1, $7, $9}' access.log | head -10

```

```

root@ubuntu:~/soc-labs/wazhu# awk '$6 == "\"POST\""' {print $1, $7, $9}' access.log | head -10
37.115.186.244 /blog/geekery/xvfb-firefox 200
78.173.140.106 /blog/geekery/pyblosxom-mdate-vim-hack.html/trackback/ 404
78.173.140.106 /blog/geekery/pyblosxom-mdate-vim-hack.html/trackback/ 404
78.173.140.106 /blog/geekery/pyblosxom-mdate-vim-hack.html/trackback/ 404
91.236.74.121 /projects/xdotool/ 200

```

13: Calculate Bandwidth Usage

```
awk '{sum += $10} END {print sum/1024/1024 " MB"}' access.log
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{sum += $10} END {print sum/1024/1024 " MB"}' access.log
2620.01 MB
```

Bandwidth by IP:

```
awk '{bytes[$1] += $10} END {for (ip in bytes) print ip, bytes[ip]/1024/1024 " MB"}' access.log | sort -k2 -rn | head -10
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{bytes[$1] += $10} END {for (ip in bytes) print ip, bytes[ip]/1024/1024 " MB"}' access.log | sort -k2 -rn | head -10
68.180.224.225 160.344 MB
94.23.164.135 155.401 MB
190.153.25.242 105.032 MB
100.2.4.116 103.636 MB
88.198.255.242 103.6 MB
184.154.149.126 103.582 MB
66.249.73.135 72.0029 MB
117.28.234.67 66.0043 MB
82.200.166.110 62.2365 MB
192.95.12.193 51.8587 MB
root@ubuntu:~/soc-labs/wazhu#
```

Part 5: Text Transformation with sed

14: Extract and Format IPs

```
awk '{print $1}' access.log | sort -u > unique_ips.txt
```

Verify:

```
wc -l unique_ips.txt
```

```
head unique_ips.txt
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{print $1}' access.log | sort -u > unique_ips.txt
root@ubuntu:~/soc-labs/wazhu# wc -l unique_ips.txt
1753 unique_ips.txt
100.2.4.116
100.43.83.137
101.119.18.35
101.199.108.50
101.226.168.196
101.226.168.198
101.226.33.222
103.245.44.13
103.247.192.5
103.25.13.22
```

15: Anonymize IP Addresses

```
sed 's/\([0-9]\{1,3\}\.\)\{3\}[0-9]\{1,3\}/\1XXX/' access.log  
| head -5
```

```
root@ubuntu:~/soc-labs/wazhu# sed 's/\([0-9]\{1,3\}\.\)\{3\}[0-9]\{1,3\}/\1XXX/' access.log | head -5  
9.XXX - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"  
9.XXX - - [17/May/2015:10:05:43 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 171717 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"  
9.XXX - - [17/May/2015:10:05:47 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 26185 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.77 Safari/537.36"
```

16: Extract URLs Only

Create a file with just the requested URLs.

```
awk '{print $7}' access.log | sort -u > urls.txt
```

Filter for specific file types:

```
awk '{print $7}' access.log | grep -E "\.(php|asp|jsp)$" | sort -u
```

```
root@ubuntu:~/soc-labs/wazhu# awk '{print $7}' access.log | sort -u > urls.txt  
root@ubuntu:~/soc-labs/wazhu# awk '{print $7}' access.log | grep -E "\.(php|asp|jsp)$" | sort -u  
/administrator/index.php  
/admin.php  
/wp-login.php  
root@ubuntu:~/soc-labs/wazhu#
```

Press the "Print Scrn" on your keyboard to take a screenshot